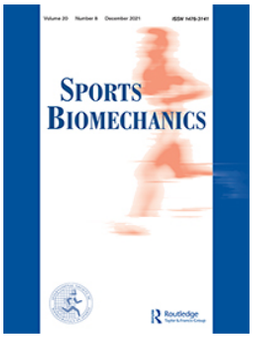

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Acute impact of Nordic hamstring exercise on sprint performance after 24, 48 and 72 hours

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ABSTRACT

The Nordic Hamstring Exercise (NHE) improves the strength of the hamstring muscles, as well as prevents and rehabilitates the injuries of said muscles. However, the eccentric demand of NHE may influence the athlete's performance, making compliance with these programmes difficult. The aim is to analyse the acute impact on sprint performance after the passing of 24, 48, and 72 hours respectively since an NHE-based session (4 sets of 10 repetitions) had taken place. Participants were randomly divided into an experimental group (EG) (n = 12 male participants) who carried out an NHE session and a measurement of their 30 m sprint performance in each of the three subsequent days, and a control group (CG) (n = 12 male participants) who did not take part in the NHE session. The results show a significant reduction of maximum power within 24 hours ($t = 3.57$, $d = 0.22$, $P < .0273$) as well of the production of high speed horizontal force up to after 48 hours ($t = 4.82$, $d = 0.22$, $P < .0001$) in the EG. These results may suggest separating weekly NHE sessions from competition or demanding training in which sprint performance should not be affected by at least 72 hours.

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Athletic performance; hamstring muscles; muscle strength; periodisation; velocity

Introduction

Sprinting capacity is a fundamental component of numerous sports activities in which performance is determined by covering a concrete distance in the least time possible (Bourne et al., 2018; Morin et al., 2011). The performance of the sprint acceleration results from the sprinters' abilities to produce and maintain on the ground a horizontally, backward-oriented force (Morin et al., 2011).

The hamstring muscles (HM) have a fundamental role in the production of horizontal force (F0), required for the relative maximum output in a horizontal direction during sprint acceleration (Morin & Samozino, 2016) which is more important at low speeds (Cavagna et al., 1971) as in the short sprints involved in team sports. The F0 closely associates with the theoretical maximal running velocity the athlete would be able to

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reach should mechanical resistances (i.e., internal and external) against movement be null (V0), it also represents the capability to produce F0 at very high running velocities (Morin & Samozino, 2016), which in turn results in a better sprint performance (Colyer et al., 2018). The F0 ability, therefore, seems to be particularly relevant for short accelerations, whilst the V0 ability would be relevant for long accelerations and for achieving a high sprint speed (Samozino et al., 2016). Furthermore, the relationship between these variables encompasses the whole capacity of the neuromuscular system, including the mechanical properties of the muscles, their morphological characteristics, and the neural mechanisms supporting the impulse of the motor unit (Jiménez-Reyes, García-Ramos et al., 2018).

HM´s contribution to horizontal force production during the support phase is determined by their ability to generate high levels of eccentric strength and to be activated at the end of the 'swing' phase (Alt, Komnik et al., 2021; Chumanov et al., 2007; Morin et al., 2015; Schache et al., 2011). This muscle group has a central role in any high-speed race (Morin et al., 2015) and sprinting places a demand on it mostly between the mid-swing phase and the foot strike or early stance (Kenneally-Dabrowski et al., 2019).

At the same time, HM seems to be the most vulnerable region in those sports, which involve high-speed racing (Edouard et al., 2016). This range of problems has allowed the scientific community to seek strategies for the prevention and limitation of the high incidence of hamstring strain injuries (HSI). Despite the effort made for prevention and treatment of these injuries (Brooks et al., 2005), their incidence has not decreased (Gabbett, 2016). Therefore, different types of exercises such as the eccentric fly wheel leg curl, nordic exercise, leg curl eccentric or stiff-leg deadlift, among others, meant to strengthen this muscle group have been suggested as a promising approach to the primary prevention of HIS (Bourne et al., 2018), although there has also been some debate as to their practical efficacy (Van Dyk et al., 2019).

Thus, within this group of exercises, the eccentric loading and the long-length exercises reduce the risk of HIS (Brockett et al., 2001), increase the strength (Timmins et al., 2016) and the hamstring muscle fascicle length in the non-injured individuals (Alonso-Fernandez et al., 2018) and when used during the period of rehabilitation it is shortened (Askling et al., 2014). One of the most important exercises of this kind is the 'Nordic Hamstring Exercise' (NHE) which is capable of bringing about this type of eccentric stimulus, proving its capacity to reduce the incidence and seriousness of hamstring injuries in both soccer (Arnason et al., 2008) and rugby (Brooks et al., 2006). The NHE is an open kinetic chain, eccentric loading, mostly peer-assisted exercise, which increases both the eccentric strength (Mjolsnes et al., 2004) and the hamstring muscle fascicle length (Alonso-Fernandez et al., 2018). Implementing a fixed resistance at the heels is favourable for stability, intensity, and safe training (Mjolsnes et al., 2004). Moreover, the force production during NHE has recently been widely correlated with force production during sprinting, both at low (F0) and at high speed (V0) (Alt, Severin et al., 2021; Ishøi, Hölmich, Aagaard, Thorborg, Bandholm, Serner et al., 2018a; Prince et al., 2021), proving itself to be an interesting exercise for the development of this ability.

Despite the proven practical transfer, NHE has also been the object of criticism, in that it is an exercise, which is difficult to adapt to the real training dynamics of athletes during the season (Cuthbert et al., 2020). This is certainly due to the fact that eccentric exercise in the case of persons lacking physical fitness results in damage to the muscle fibre and to delayed onset muscle soreness (DOMS) in the following days (Brockett et al., 2001) which may have a negative influence on performance. It has been noted that the high eccentric load of NHE has had a negative acute impact on sprint performance, leading to a decrease in the capacity of repeating sprints after performing NHE (Timmins et al., 2014). This is a circumstance that could increase the acute risk of HSI.

Traditionally, the weekly application of NHE was based on high numbers of repetitions of the exercise (Mjolsnes et al., 2004). However, it has recently been noted that positive effects could be obtained even with a lower number of repetitions during each week (Alt, Knicker et al., 2021; Cuthbert et al., 2020; Pollard et al., 2019; Presland et al., 2018). It would, therefore, seem that the effect might be due to the intensity with which NHE is performed (Alt, Knicker et al., 2021; Pollard et al., 2019; Presland et al., 2018). This intensity should grow progressively, just as in the case of traditional strength training (Cuthbert et al., 2020), e.g., by reducing external assistance like reported by Alt et al. Presland et al. (2018), Alt et al. (2018) considering also the latter stages of the NHE are the most demanding but the most important for performance enhancement and injury prevention as well (Alt, Knicker et al., 2021; Alt et al., 2018). The problem consists of structuring the intensity in such a way as to not have a negative influence on sports performance and on keeping to the schedule (which could be affected by the athlete's experiencing DOMS).

In summary, hamstring muscles play an important role in the production of horizontal force during sprinting (Morin et al., 2015), while sprinting is an important cause of HIS (Edouard et al., 2016). The eccentric hamstring strength gain increases the production of F0 and V0 in sprint performance (Morin et al., 2015) as well as the capacity to maintain its absolute levels during fatigue (Timmins et al., 2014), thus reducing the risk of suffering HIS (Timmins et al., 2016). Therefore, the optimal adjustment of an eccentric loading such as NHE may improve sprint performance as well as prevent HIS (Ishøi, Hölmich, Aagaard, Thorborg, Bandholm, Serner et al., 2018b).

NHE proved to be an interesting exercise in the prevention of HSI, but the protocols used in its implementation were different. It would be important to investigate the timing of NHE training within a microcycle/training week so that it does not interfere with sprints/acceleration at highest effort. And thus, to find out how an NHE-based acute eccentric impact influences sprint performance on successive days, in order to obtain a window of time, which would allow coaches to schedule its appropriate weekly dosage in the initial stages.

As a result, the aim of the present study was to observe the acute impact of an NHE session on a 30 m sprint performance following a period of 24, 48, and 72 hours respectively from its implementation in individuals without prior experience. It is hypothesised that, after applying an NHE session, sprint performance will be negatively influenced on subsequent days.

Methods

Sample

The sample consisted of 24 male individuals (age: 22.5 ± 2.6 years, height: 1.77 ± 0.11 m, body mass: 75.7 ± 10.3 kg). All participants were physically active, with a regular practice (>5 sessions per week) of competitive sports (team sports, martial arts, and athletics) at different levels. All participants were familiar with the sprint technique due to their current or previous sports practice. The selection criteria were not to have suffered from any HSI in the 12 previous months and not to have had any previous experience with eccentric NHE training. All participants were fully informed of the features of the study and accepted to participate of their own volition, signing the required informed consent. All the measures and gathering of data took place at the Faculty of Education and Sport Sciences, University of Vigo (Pontevedra, Spain). The study was approved by the Autonomous Ethics Committee of Research of Xunta de Galicia, Ministry of Health (Spain) (registration no.: 2018/239) and all procedures were designed and administered in accordance with the Helsinki Declaration.

Design

The study was carried out over a period of 3 weeks. In weeks 1 and 2, the participants had four sessions of conditioning for sprint in order to improve their technique and performance in a 30 m trial chosen for its optimal connection to performance in a variety of sports. Besides, during week 1 there were two sessions in which the participants became acquainted with NHE in order to guarantee both technical quality and the proper intensity during its performance. In these sessions, the participants learned to execute the movement in a controlled and constant a manner, without noticeably modifying the angle of the hip during the descent phase, performing three series of 5 repetitions in each session.

During week 3, the experimental stage was carried out. A first measurement session (M1) was carried out, preceded by a 15-minute standardised warm-up consisting of continuous jogging, dynamic stretching and sprint technique. M1 consisted of anthropometric evaluation (height and body mass) of each subject and of their performance in a 30 m sprint. Once those measurements were completed, the participants were randomly assigned to two groups: experimental (EG = 12) and control group (CG = 12). After this assignation, the participants of the EG performed a complete NHE training session, consisting of 4 series of 10 repetitions each, separated by a 3 minute break (Alt, Knicker et al., 2021; Alt et al., 2018; Presland et al., 2018).

We intented to apply a type of session based on the general characteristics described in the previous literature. Therefore, the decision to use this design was based on the systematic review of Cuthbert et al. (2020), in which the average number of weekly repetitions of NHE training at the end of each intervention averaged 51.36 ± 17.7 repetitions per week. Consequently, as this was an analysis of the acute effect, we decided to use 4 sets of 10 repetitions to find an average of the repetitions used in the works of the aforementioned systematic review. With the same reasoning, it was

decided that the type of NHE used should be partner-assisted and without additional equipment. Although there are other, more controlled ways to apply it (Alt, Knicker et al., 2021; Alt et al., 2018; Presland et al., 2018), this is how it is usually used in real sports training on most occasions, so we would contribute to guarantee its ecological validity. The NHE is a supra-maximal eccentric exercise, interesting for practical purposes because of its simplicity and low equipment needs. Keeping this perspective is what led us to choose our experimental design.

The measurements of sprint performance following the standardised warm-up were repeated for all participants (EG and CG) at 24 hours (M2), 48 hours (M3), and 72 hours (M4) after M1. In addition, all subjects stated their respective perceived levels of recovery and soreness following each warm-up and before every measurement session, using a visual analogue scale (Alt et al., 2018). All measurements were carried out at the same hour of the day and in identical conditions in order to minimise any bias.

Sprint performance

It was measured with a manual radar (Stalker ATS II, Minneapolis), placed on a 2 m tripod behind the subjects initial position and at a height of 1 m, approximately corresponding to the subject's centre of mass (Prince et al., 2021). Using the data obtained by the radar in each sprint in order to establish the strength–speed relationships, the following were calculated by means of linear regression (Morin et al., 2019; Samozino et al., 2016): the horizontal force production at low speed (F0), the horizontal force production at high speed (V0), the horizontal power (Pmax), the rate of force application to sprint (RFmax), the rate of force reduction during sprint (Drf), and the maximum speed achieved (MaxSpeed). During each measurement session (M1-M2-M3-M4) on indoor athletics track, each subject performed three 30 m sprints with two-minute breaks between them and preceded by a standardised warm-up. The recorded value was the average of the three sprints, since a better reliability of the measurements can be achieved by tracing the average of two or three sprints, rather than the best of the three sprints (Simperingham et al., 2019; Mendiguchia et al. 2014).

Nordic hamstring exercise

The EG subjects carried out a complete NHE session at the end of the first measurement session (M1). Participants performed only the eccentric phase of the NHE (to decrease fatigue and increase eccentric intensity), with the assistance of a partner and on a lightly padded surface to encourage natural patellar motion to avoid joint stress that can cause patellofemoral pain (Alt et al., 2018) (Figure 1). Minimal hip flexion was used to maximise the lever arm of the knee joint centre of mass to increase the eccentric load on the knee flexors. The participants were asked to meet the following execution quality criteria: (a) to execute eccentric NHEs with a constant knee extension velocity of ~15°/s across the largest possible knee joint ROM (~90°) targeting a time under tension of ~6 seconds per repetition (using a metronome) and (b) to realise minimal hip flexion whilst (c) the hands were positioned in front of the shoulders (Alt, Knicker et al., 2021; Alt et al., 2018).

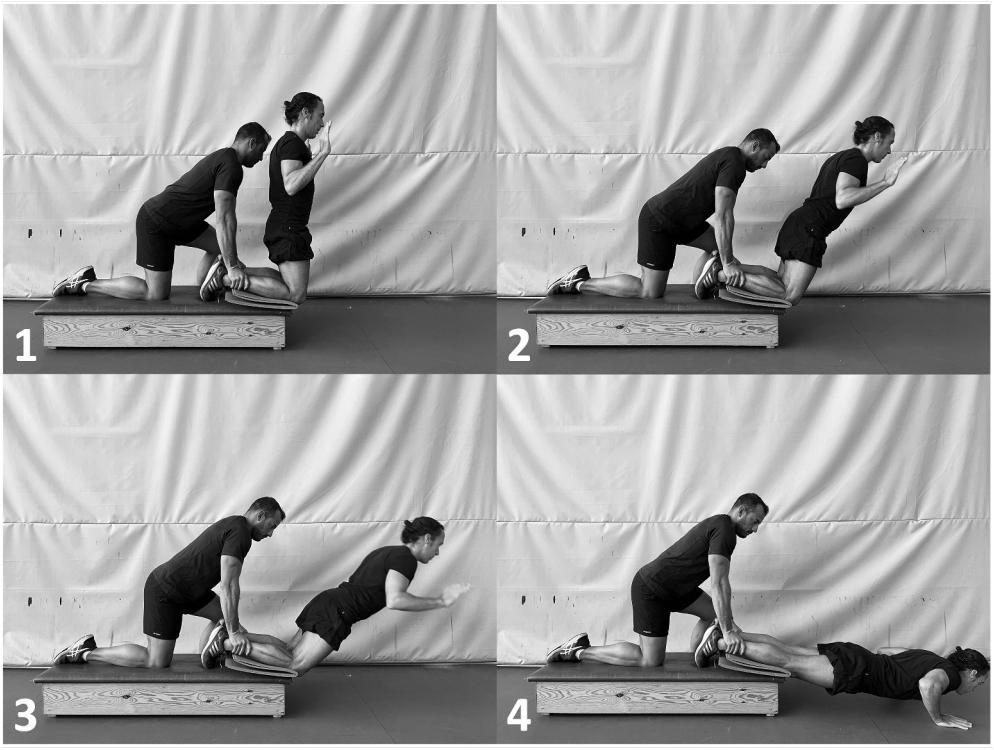

Figure 1. Nordic hamstring exercise: (1) start, (2,3) intermediate points, (4) end point. The NHE was performed on an elevated surface (12 cm) to maximise the knee range motion (~90°) and the time under tension of the eccentric movement (~6s). A padded support was used under the shins until tibial tuberosity, with the knees on edge to free the patella and ensure its natural movement (Alt et al., 2018).

Perceived soreness and recovery

The EG subjects were consulted prior to every measurement session regarding their levels of perceived recovery and muscle soreness by means of a subjective evaluation scale ranging from 1 (complete recovery and absence of muscle soreness) to 10 (total fatigue and debilitating muscle pain) (Figure 2).

Statistical analysis

The normal distribution of data was checked by means of the Shapiro–Wilk test and homoscedasticity through Levene's test. The Greenhouse–Geisser correction was used when the test of sphericity was violated ($P < .05$ for Mauchly's test of sphericity). An ANOVA repeated measurement design was used in order to ascertain the changes induced by NHE in sprint performance (F0, V0, Pmax, RFmax, Drf, and MaxSpeed). The within-subjects variable was the time (M1-M2-M3-M4), and the one between-subjects was the group (EG, CG). In addition, *post-hoc* tests were used, along with the Bonferroni correction in order to establish the pairwise comparisons between measured variables (M1-M2, M1-M3, and

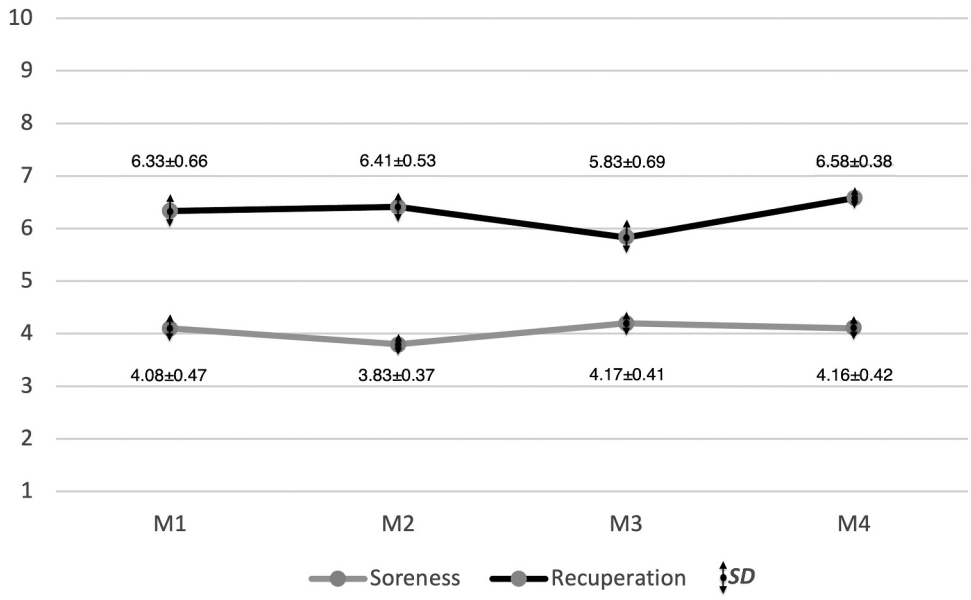

Figure 2. Mean daily soreness and recuperation measured using a numeric pain rating scale (1–10) at the beginning of each session for EG (mean ± standard deviation).

M1-M4). The significance was set at P < .05, and, where appropriate, Cohen's d was used to establish the measure of effect size. The analyses were performed using the SPSS software, version 25.0 for MacOS (IBM Corporation, Chicago, IL).

Results

No significant differences were observed between the groups in matters of age, height, or body mass (P > 0.05) (Table 1). The compliance rate (attendance and participation on the training and assessment sessions) was excellent for both the EG (99.5%) and the CG (100%).

Table 2 shows the results obtained on the basis of different variables of sprint performance measured in the sample.

Sprint performance

In EG significant *group×time* interaction was observed for V0 ($F(3–33) = 4.045$, $P = .0243$, $n_p^2=0.269$), Pmax ($F(3–33) = 4.467$, $P = .0226$, $n_p^2=0.289$), and MaxSpeed ($F(1.44–15.84) = 8.332$, $P = .0191$, $n_p^2=0.431$). However, neither F0 ($F(3–33) = 1.44$, $P > .05$,

Table 1. Characteristics of the participants; mean ($\bar{x}$); standard deviation (SD).

		Age (years old)		Body mass (kg)		Height (m)	
Group	N	($\bar{x}$)	(*SD*)	($\bar{x}$)	(*SD*)	()	()
EG	12	22.1	2.1	74.6	9.8	1.78	0.11
CG	12	22.8	3.2	76.8	10.9	1.79	0.11

EG, Experimental Group; **CG**, Control Group

Table 2. Changes in the variables in the EG and CG between baseline (M1) and 24 h (M2), 48 h (M3), and 72 h (M4) after (*mean ± standard deviation*).

	M1 (baseline)	M2 (+24 h)	% M1vsM2	M3 (+48 h)	% M1vsM3	M4 (+72 h)	% M1vsM4
EG							
F0 (N/kg)	5.31 ± 0.62	5.10 ± 0.59	−3.95	5.29 ± 0.58	−0.38	5.22 ± 0.62	−1.69
V0 (m/s)	8.91 ± 1.30	8.82 ± 1.31	−1.01	8.61 ± 1.27**	−3.37 *d* = 0.22	8.66 ± 1.34	−2.81
Pmax (W/kg)	11.90 ± 2.65	11.30 ± 2.59*	−5.04 *d* = 0.22	11.45 ± 2.46	−3.78	11.63 ± 2.44	−2.27
RFmax (%)	36.78 ± 3.62	35.79 ± 3.56	−2.69	36.42 ± 3.45	−0.98	35.90 ± 3.43	−2.39
Drf (%)	−5.76 ± 0.82	−5.61 ± 0.69	−2.60	−5.97 ± 0.91	3.65	−5.93 ± 0.97	2.95
MaxSpeed (m/s)	8.11 ± 0.93	7.95 ± 0.94*	−1.97 *d* = 0.56	7.85 ± 0.95**	−3.21 *d* = 0.93	8.08 ± 0.97	−0.37
CG							
F0 (N/kg)	5.14 ± 0.54	5.12 ± 0.72	−0.39	5.08 ± 0.90	−1.17	5.13 ± 0.98	−0.19
V0 (m/s)	8.59 ± 1.19	8.60 ± 1.06	0.12	8.44 ± 1.21	−1.75	8.46 ± 1.12	−1.51
Pmax (W/kg)	11.14 ± 1.79	11.02 ± 1.97	−1.08	10.81 ± 2.62	−2.96	10.95 ± 2.71	−1.71
RFmax (%)	35.78 ± 2.86	35.49 ± 3.15	−0.81	35.17 ± 4.74	−1.70	35.31 ± 4.97	−1.31
Drf (%)	−5.74 ± 0.78	−5.76 ± 0.98	0.35	−5.81 ± 1.04	1.22	−5.83 ± 1.02	1.57
MaxSpeed (m/s)	7.89 ± 0.71	7.83 ± 0.73	−0.76	7.71 ± 0.93	−2.28	7.73 ± 0.87	−2.03

* = $P < .05$ vs M1, ** = $P < .001$ vs M1, *d* = effect size // **F0**, horizontal force low speed; **V0**, horizontal force high speed; **RFmax**, % force application sprint; **Pmax**, horizontal power; **Drf**, % force reduction during sprint; **MaxSpeed**, max speed archived.

n_p^2=0.116), nor RFmax (F(3–33) = 12.13, $P > .05$, n_p^2=0.162), nor Drf (F(3–33) = 1.972, $P > .05$, n_p^2=0.152) were affected by NHE. In CG, no significant interactions between *group×time* were observed for any of the variables: F0 (F(2.1–22.9) = 0.047, $P > .05$, n_p^2=0.004), V0 (F(2.4–26.8) = 0.638, $P > .05$, n_p^2=0.055), Pmax (F(2.3–25.4) = 0.065, $P > .05$, n_p^2=0.024), RFmax (F(1.9–21.7) = 0.213, $P > .05$, n_p^2=0.019), Drf (F(3–33) = 0.079, $P > .05$, n_p^2=0.007), MaxSpeed (F(2.1–23.6) = 0.372, $P > .05$, n_p^2=0.111). Likewise, no significant differences were observed between EG and CG at any measurement point (M1-M2-M3-M4) and in any variable (all $P > .05$).

Regarding the pairwise comparisons derived from the *post-hoc* analysis of V0, we observed that, in the EG, there was a significant decrease with the elapse of 48 h from baseline (t = 4.82, d = 0.22, P = .0001) and there were no changes with regard to it after 24 (t = 0.82, $P > .05$) or 72 h (t = 2.55, $P > .05$). The CG showed no significant differences after either 24 (t = −0.07, $P > .05$), 48 (t = 0.8, $P > .05$), or 72 h (t = 0.83, $P > .05$) with regard to baseline.

In the Pmax pairwise comparisons, we noted that, in the EG, there was a significant decrease with the elapse of 24 h from baseline (t = 3.57, d = 0.22, P = .0273), and there were no changes with regard to it after 48 (t = 2.26, $P > .05$) or 72 h (t = 3.13, $P > .05$). The CG showed no significant differences after either 24 (t = 0.49, $P > .05$), 48 (t = 0.77, $P > .05$) or 72 h (t = 0.41, $P > .05$) with regard to baseline.

Regarding MaxSpeed, we observed that in the EG there was a significant difference after 24 (t = 4.22, d = 0.56, P = .0091) and 48 h (t = 6.42, d = 0.93, P = .0001) from the baseline values, and there were no changes with regard to it after 72 h (t = 1.24, $P > .05$). The CG showed no significant differences after either 24 (t = 0.9, $P > .05$), 48 (t = 1.48, $P > .05$) or 72 h (t = 1.43, $P > .05$) with regard to baseline.

In the pairwise comparisons of F0, no significant changes were observed, in either the EG (t = 1.79, P > .05) or the CG (t = 0.16, P > .05) after 24, 48 (EG t = 0.15, P > .05, CG t = 0.29, P > .05) or 72 h (EG t = 0.79, P > .05, CG t = 0.04, P > .05) from baseline.

Similar results were observed in RFmax comparisons. Neither the EG (t = 2.33, P > .05) nor the CG (t = 0.49, P > .05) show significant changes after 24, 48 (EG t = 0.85, P > .05, CG t = 0.63, P > .05) or 72 h (EG t = 2.05, P > .05, CG t = 0.48, P > .05) from baseline.

Likewise, in the analysis of Drf, neither the EG (t = −0.97, P > .05) nor the CG (t = 0.11, P > .05) showed significant changes after 24, 48 (EG t = 0.85, P > .05, CG t = 0.29, P > .05) or 72 h (EG t = 2.05, P > .05, CG t = 0.37, P > .05) from baseline.

Soreness and perceived recovery

During the experimental stage, neither perceived soreness (F(1.94–21.38) = 0.204, P > .05, n_p^2=0.018) nor perceived recovery (F(2–22) = 0.537, P > .05, n_p^2=0.047) were affected by NHE. The average soreness measurement reported over the days was 4.06 ± 1.44 and that of perceived recovery 6.28 ± 0.58 (±$S\bar{D}$).

Discussion and implications

The main findings of this study were that (1) the horizontal force production at high speed (V0) and the achieved maximum speed (MaxSpeed) were significantly diminished up to 48 h after the application of an NHE-based acute eccentric stimulus, recovering its values after 72 h. However, (2) the horizontal force production at low speed (F0) as well as the subjective perception of soreness and recovery did not seem to be influenced by said initial stimulus. Therefore, the hypothesis that sprint performance is temporarily influenced by an NHE session is confirmed for some of the performance variables studied. To the best of the authors' knowledge, this is the first study on the acute impact on sprint performance at 72 h after an NHE-based intervention session.

Our results showed a change in sprint performance 48 h after being exposed to an NHE-based acute eccentric impact. More specifically, said stimulus did not alter the capacity of producing horizontal force at low speed (F0), but it did significantly alter the application of maximum power (Pmax) after 24 h and the capacity of producing horizontal force at high speed (V0) after 48 h. These effects on sprint are in line with the results obtained by Jiménez-Reyes et al. (2018) showing that the capacity of applying horizontal force at high speeds was altered, contrary to the data of the initial acceleration.

This change of the force production at high speed could be important not only for the decrease in muscle performance, but also due to a possible negative influence in the quality of the specific sporting abilities (Morin et al., 2011). Thus, one could speculate that the decreases in V0, greater than those in F0 could have been the result of the decrease in the capacity of energy generation of the selected fibres. This fact should be due to greater alterations induced by metabolism (Bishop & Spencer, 2004), especially FT fibres because eccentric NHE training selectively promotes explosive strength capacitites (Alt et al., 2018). On the other hand, the lesser decrease in F0 could be due to the fact that the PCr re-synthesis ways recovered relatively quickly and the first seconds of effort were

not affected by fatigue as much as the following ones; this could be backed by the deterioration of the sarcomere excitability (Perrey et al., 2010). It seems, therefore, that V0 and its support mechanisms are more sensitive to fatigue than F0, but also to the fact that, during a sprint, the production of force at high speed always took place after 2 or 3 seconds of effort and, as a result, the athletes were more exposed to fatigue than the capacity of force production at low speed (F0) which occurred in the first seconds (Jiménez-Reyes, Cross et al., 2018).

However, we should not dismiss that another explanation could be the possible muscle damage derived from the NHE and the execution of the sprints. It has been demonstrated that muscle damage 2–7 days after eccentric knee flexion was prominent in the semitendinosus as compared with the other hamstring muscles (Kubota et al., 2007). Meanwhile, and although in longer sprints than those in the present study, semitendinosus muscle volume normalised to body mass correlated significantly with sprint speed in the 50–60 m interval, the volume of the other hamstring muscles did not (Takahashi et al., 2020). Consequently, considering these findings, it is possible that the observed decreases in MaxSpeed and V0 were mainly produced by the damage of the semitendinosus.

The compliance in terms of the real-world application of NHE interventions, or of the programs meant to deal with HM injuries, is rather low (Ishøi, Hölmich, Aagaard, Thorborg, Bandholm, Serner et al., 2018c). Even if this fact lacks a concrete justification, a common reason for which athletes do not comply with NHE interventions—or coaches do not prescribe NHE—is the level of DOMS experienced as a consequence of the big eccentric stimulus provided by this exercise (Cuthbert et al., 2020).

Previous studies (Alt et al., 2018; Freeman et al., 2019; Mjolsnes et al., 2004) identified DOMS or muscular pain by means of a subjective analysis scale after each warm-up preceding the NHE intervention session. The recorded scores were not higher than six on a scale from one to ten, i.e., levels of soreness being between moderate and low. These data support the results of the present study, obtained with a similar instrument, where both soreness and perceived recovery stayed within moderate and consistent values (4.06 ± 1.44 and 6.28 ± 0.58 respectively), guaranteeing a high rate of compliance (99.5%). However, in the study conducted by Ishøi, Hölmich, Aagaard, Thorborg, Bandholm, Serner et al., 2018c) the subjects who reported HM soreness, which would have been the reason for a low compliance with the protocol, were eliminated from the study. One must consider that the level of soreness as well as the difference between discomfort and soreness are individual and subjective measures, which may constitute a decisive factor when it comes to complying with the sessions of a protocol.

Our perceived soreness data do not support the results of Ishøi, Hölmich, Aagaard, Thorborg, Bandholm, Serner et al., 2018c), but we should take into account that it was only one NHE session, rather than a whole protocol, although in the initial sessions there was, habitually, a smaller muscle conditioning and, hence, higher levels of perceived soreness. Therefore, one would have to observe the evolution of this variable more closely in order to understand whether this is truly a limiting factor in the use of NHE protocols and in sprint performance.

In practical terms, given this perspective and with the data obtained in the present study, we can conclude that it seems reasonable to leave a 72 h recovery period between a high NHE-based eccentric impact and other training or competition sessions in which a high level of motor performance is required, at least until the subjects are completely conditioned for the exercise. This would have to be taken into account especially when the training sessions or competitions ahead involve the application of horizontal force to a large extent. Moreover, we cannot lose sight of the fact that a reduced performance in sprint could also increase the risk of HIS (Timmins et al., 2016).

In our case, 72 h were needed in order to observe that the totality of the sprint-related variables we studied recovered values similar to the initial ones and therefore neither hampered performance, nor increased the potential HSI risk. This could prove important for the structuring of weekly sessions centred on NHE, scheduling them well before or after competitions or quality training. However, when it comes to another type of protocols, such as the rehabilitation ones, in which the athletes would not be conditioned by their performance, its weekly frequency may be higher.

Nevertheless, the authors are aware of the limitations of the present study. Recent scientific literature (Alt, Knicker et al., 2021; Pollard et al., 2019; Presland et al., 2018). Points out the possibilities of using lower volumes of NHE, so that applying a session model with a more contained volume of series/repetitions could influence the results obtained. Likewise, having only a subjective measure of pain/recovery of the EG limits our interpretation. Having a direct measure, such a magnetic resonance image, of hamstring muscle damage for both groups would improve our interpretation of the results and how the performance of daily sprints may additionally influence fatigue. Another area for improvement would be the use of kinematic analysis to control the intensity of NHE execution (Alt et al., 2018), although its implementation in real sports training situations could be difficult. Finally, it should be remembered that this study involved participants with no previous experience in eccentric training so that in the future it would be interesting to compare the results obtained in other groups with a greater degree of previous experience. In this way, when training with inexperienced and/or weaker individuals, maybe a NHE-assisted execution (Alt, Knicker et al., 2021; Marshall et al., 2015) could be a good strategy to increase the intensity with fewer repetitions, achieving a higher time under tension at extended knee angles. Especially in the last repetitions of the NHE without assistance, where in some participants of the present study a decrease in time under tension due to fatigue could be observed.

The present study represents no more than a step towards improving the structuring of the use of NHE in HSI prevention and rehabilitation protocols. It indicates the impact that this exercise could have on sprint performance, subsequent sporting performance and, paradoxically, on the increased risk of HSI by an increasing fatigue that weakens the hamstrings (Heiderscheit et al., 2010). Further studies will be needed in order to determine the correct medium-term administration of this stimulus; however, we were able to observe that, in the short term, its use should also be correctly temporarily periodised, until such point as the athletes are adequately conditioned.

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References

Alonso-Fernandez, D., Docampo-Blanco, P., & Martinez-Fernandez, J. (2018, January). Changes in muscle architecture of biceps femoris induced by eccentric strength training with nordic hamstring exercise. *Scandinavian Journal of Medicine & Science in Sports*, *28*(1), 88–94. https://doi.org/10.1111/sms.12877

Alt, T., Knicker, A. J., Nodler, Y. T., & Strüder, H. K. (2021, March 16). Assisted or unassisted Nordic hamstring exercise? - Resistance exercise determinants at a glance. *Sports Biomechanics*, 1–15. https://doi.org/10.1080/14763141.2021.1893376

Alt, T., Komnik, I., Severin, J., Nodler, Y. T., Benker, R., Knicker, A. J., Brüggemann, G. P., & Strüder, H. K. (2021, January 13). Swing phase mechanics of maximal velocity sprints-does isokinetic lower-limb muscle strength matter? *International Journal of Sports Physiology and Performance*, *16*(7), 1–11. https://doi.org/10.1123/ijspp.2020-0423

Alt, T., Nodler, Y. T., Severin, J., Knicker, A. J., & Strüder, H. K. (2018, January). Velocity-specific and time-dependent adaptations following a standardized Nordic hamstring exercise training. *Scandinavian Journal of Medicine & Science in Sports*, *28*(1), 65–76. https://doi.org/10.1111/sms.12868

Alt, T., Severin, J., Komnik, I., Nodler, Y. T., Benker, R., Knicker, A. J., Brüggemann, G. P., & Strüder, H. K. (2021, April). Nordic hamstring exercise training induces improved lower-limb swing phase mechanics and sustained strength preservation in sprinters. *Scandinavian Journal of Medicine & Science in Sports*, *31*(4), 826–838. https://doi.org/10.1111/sms.13909

Arnason, A., Andersen, T. E., Holme, I., Engebretsen, L., & Bahr, R. (2008). Prevention of hamstring strains in elite soccer: An intervention study: Prevention of hamstring strains in soccer. *Scandinavian Journal of Medicine & Science in Sports*, *18*(1), 40–48. https://doi.org/10.1111/j.1600-0838.2006.00634.x

Askling, C. M., Tengvar, M., Tarassova, O., & Thorstensson, A. (2014). Acute hamstring injuries in Swedish elite sprinters and jumpers: A prospective randomised controlled clinical trial comparing two rehabilitation protocols. *British Journal of Sports Medicine*, *48*(7), 532–539. https://doi.org/10.1136/bjsports-2013-093214

Bishop, D., & Spencer, M. (2004). Determinants of repeated-sprint ability in well-trained team-sport athletes and endurance-trained athletes. *The Journal of Sports Medicine and Physical Fitness*, *44*(1), 1–7.

Bourne, M. N., Timmins, R. G., Opar, D. A., Pizzari, T., Ruddy, J. D., Sims, C., Williams, M. D., & Shield, A. J. (2018). An evidence-based framework for strengthening exercises to prevent hamstring injury. *Sports Medicine*, *48*(2), 251–267. https://doi.org/10.1007/s40279-017-0796-x

Brockett, C. L., Morgan, D. L., & Proske, U. (2001). Human hamstring muscles adapt to eccentric exercise by changing optimum length. *Medicine and Science in Sports and Exercise*, *33*(5), 783–790. https://doi.org/10.1097/00005768-200105000-00017

Brooks, J. H. M., Fuller, C. W., Kemp, S. P. T., & Reddin, D. B. (2005). Epidemiology of injuries in English professional rugby Union: Part 1 match injuries. *British Journal of Sports Medicine*, *39* (10), 757–767. https://doi.org/10.1136/bjsm.2005.018135

Brooks, J. H. M., Fuller, C. W., Kemp, S. P. T., & Reddin, D. B. (2006). Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *The American Journal of Sports Medicine*, *34*(8), 1297–1306. https://doi.org/10.1177/0363546505286022

Cavagna, G. A., Komarek, L., & Mazzoleni, S. (1971). The mechanics of sprint running. *The Journal of Physiology*, *217*(3), 709–721. https://doi.org/10.1113/jphysiol.1971.sp009595

Chumanov, E. S., Heiderscheit, B. C., & Thelen, D. G. (2007). The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of sprinting. *Journal of Biomechanics*, *40*(16), 3555–3562. https://doi.org/10.1016/j.jbiomech.2007.05.026

Colyer, S. L., Nagahara, R., Takai, Y., & Salo, A. I. T. (2018). How sprinters accelerate beyond the velocity plateau of soccer players: Waveform analysis of ground reaction forces. *Scandinavian Journal of Medicine & Science in Sports*, *28*(12), 2527–2535. https://doi.org/10.1111/sms.13302

Cuthbert, M., Ripley, N., McMahon, J. J., Evans, M., Haff, G. G., & Comfort, P. (2020). The effect of Nordic hamstring exercise intervention volume on eccentric strength and muscle architecture adaptations: A systematic review and meta-analyses. *Sports Medicine*, *50*(1), 83–99. https://doi.org/10.1007/s40279-019-01178-7

Edouard, P., Branco, P., & Alonso, J. M. (2016). Muscle injury is the principal injury type and hamstring muscle injury is the first injury diagnosis during top- level international athletics championships between 2007 and 2015. *British Journal of Sports Medicine*, *50*(10), 619–630. https://doi.org/10.1136/bjsports-2015-095559

Freeman, B. W., Young, W. B., Talpey, S. W., Smyth, A. M., Pane, C. L., & Carlon, T. A. (2019). The effects of sprint training and the Nordic hamstring exercise on eccentric hamstring strength and sprint performance in adolescent athletes. *The Journal of Sports Medicine and Physical Fitness*, *59*(7), 1119–1125. https://doi.org/10.23736/S0022-4707.18.08703-0

Gabbett, T. J. (2016). The training—injury prevention paradox: Should athletes be training smarter and harder? *British Journal of Sports Medicine*, *50*(5), 273–280. https://doi.org/10.1136/bjsports-2015-095788

Heiderscheit, B. C., Sherry, M. A., Silder, A., Chumanov, E. S., & Thelen, D. G. (2010). Hamstring strain injuries: Recommendations for diagnosis, rehabilitation, and injury prevention. *Journal of Orthopaedic & Sports Physical Therapy*, *40*(2), 67–81. https://doi.org/10.2519/jospt.2010.3047

Ishøi, L., Hölmich, P., Aagaard, P., Thorborg, K., Bandholm, T., & Serner, A. (2018a). Effects of the Nordic hamstring exercise on sprint capacity in male football players: A randomized controlled trial. *Journal of Sports Sciences*, *36*(14), 1663–1672. https://doi.org/10.1080/02640414.2017.1409609

Ishøi, L., Hölmich, P., Aagaard, P., Thorborg, K., Bandholm, T., & Serner, A. (2018b). Effects of the Nordic hamstring exercise on sprint capacity in male football players: A randomized controlled trial. *Journal of Sports Sciences*, *36*(14), 1663–1672. https://doi.org/10.1080/02640414.2017.1409609

Ishøi, L., Hölmich, P., Aagaard, P., Thorborg, K., Bandholm, T., & Serner, A. (2018c). Effects of the Nordic hamstring exercise on sprint capacity in male football players: A randomized controlled trial. *Journal of Sports Sciences*, *36*(14), 1663–1672. https://doi.org/10.1080/02640414.2017.1409609

Jiménez-Reyes, P., Cross, M., Ross, A., Samozino, P., Brughelli, M., Gill, N., & Morin, J.-B. (2018). Changes in mechanical properties of sprinting during repeated sprint in elite rugby sevens athletes. *European Journal of Sport Science*, *19*(5), 585–594. https://doi.org/10.1080/17461391.2018.1542032

Jiménez-Reyes, P., García-Ramos, A., Cuadrado-Peñafiel, V., Párraga-Montilla, J. A., Morcillo-Losa, J. A., Samozino, P., & Morin, J.-B. (2018). Differences in sprint mechanical force–velocity profile between trained soccer and futsal players. *International Journal of Sports Physiology and Performance*, *14*(4), 478–485. https://doi.org/10.1123/ijspp.2018-0402

Kenneally-Dabrowski, C. J. B., Brown, N. A. T., Lai, A. K. M., Perriman, D., Spratford, W., & Serpell, B. G. (2019). Late swing or early stance? A narrative review of hamstring injury mechanisms during high-speed running. *Scandinavian Journal of Medicine & Science in Sports*, *29*(8), 1083–1091. https://doi.org/10.1111/sms.13437

Kubota, J., Ono, T., Araki, M., Torii, S., Okuwaki, T., & Fukubayashi, T. (2007). Non-uniform changes in magnetic resonance measurements of the semitendinosus muscle following intensive eccentric exercise. *European Journal of Applied Physiology*, *101*(6), 713–720. https://doi.org/10.1007/s00421-007-0549-x

Marshall, P. W., Lovell, R., Knox, M. F., Brennan, S. L., & Siegler, J. C. (2015). Hamstring fatigue and muscle activation changes during six sets of Nordic hamstring exercise in Amateur Soccer Players. *Journal of Strength and Conditioning Research*, *29*(11), 3124–3133. https://doi.org/10.1519/JSC.0000000000000966

Mendiguchia, J., Samozino, P., Martinez-Ruiz, E., Brughelli, M., Schmikli, S., Morin, J. B., and Mendez-Villanueva, A. (2014, July). Progression of mechanical properties during on-field sprint running after returning to sports from a hamstring muscle injury in soccer players. *International Journal of Sports Medicine*, *35* (8), 690–695. https://doi.org/10.1055/s-0033-1363192

Mjolsnes, R., Arnason, A., Østhagen, T., Raastad, T., & Bahr, R. (2004). A 10-week randomized trial comparing eccentric vs. Concentric hamstring strength training in well-trained soccer players. *Scandinavian Journal of Medicine and Science in Sports*, *14*(5), 311–317. https://doi.org/10.1046/j.1600-0838.2003.367.x

Morin, J. B., Edouard, P., & Samozino, P. (2011). Technical ability of force application as a determinant factor of sprint performance. *Medicine & Science in Sports & Exercise*, *43*(9), 1680–1688. https://doi.org/10.1249/MSS.0b013e318216ea37

Morin, J. B., Gimenez, P., Edouard, P., Arnal, P., Jiménez-Reyes, P., Samozino, P., Brughelli, M., & Mendiguchia, J. (2015). Sprint acceleration mechanics: The major role of hamstrings in horizontal force production. *Frontiers in Physiology*, *24*(6), 404. https://doi.org/10.3389/fphys.2015.00404

Morin, J. B., & Samozino, P. (2016). Interpreting power-force-velocity profiles for individualized and specific training. *International Journal of Sports Physiology and Performance*, *11*(2), 267–272. https://doi.org/10.1123/ijspp.2015-0638

Morin, J. B., Samozino, P., Murata, M., Cross, M., & Nagahara, R. (2019). A simple method for computing sprint acceleration kinetics from running velocity data: Replication study with improved design. *Journal of Biomechanics*, *94*, 82–87. https://doi.org/10.1016/j.jbiomech.2019.07.020

Perrey, S., Racinais, S., Saimouaa, K., & Girard, O. (2010). Neural and muscular adjustments following repeated running sprints. *European Journal of Applied Physiology*, *109*(6), 1027–1036. https://doi.org/10.1007/s00421-010-1445-3

Pollard, C. W., Opar, D. A., Williams, M. D., Bourne, M. N., & Timmins, R. G. (2019, May 5). Razor hamstring curl and Nordic hamstring exercise architectural adaptations: Impact of exercise selection and intensity. *Scandinavian Journal of Medicine & Science in Sports*, *29*(5), 706–715. https://doi.org/10.1111/sms.13381

Presland, J. D., Timmins, R. G., Bourne, M. N., Williams, M. D., & Opar, D. A. (2018, July). The effect of Nordic hamstring exercise training volume on biceps femoris long head architectural adaptation. *Scandinavian Journal of Medicine & Science in Sports*, *28*(7), 1775–1783. https://doi.org/10.1111/sms.13085

Prince, C., Morin, J. B., Mendiguchia, J., Lahti, J., Guex, K., Edouard, P., & Samozino, P. (2021). Sprint specificity of isolated hamstring-strengthening exercises in terms of muscle activity and force production. *Frontiers in Sports and Active Living*, *2*, 609636. https://doi.org/10.3389/fspor.2020.609636

Samozino, P., Rabita, G., Dorel, S., Slawinski, J., Peyrot, N., Saez De Villarreal, E., & Morin, J.-B. (2016). A simple method for measuring power, force, velocity properties, and mechanical effectiveness in sprint running: Simple method to compute sprint mechanics. *Scandinavian Journal of Medicine & Science in Sports*, *26*(6), 648–658. https://doi.org/10.1111/sms.12490

Schache, A. G., Blanch, P. D., Dorn, T. W., Brown, N. A., Rosemond, D., & Pandy, M. G. (2011, July). Effect of running speed on lower limb joint kinetics. *Medicine & Science in Sports & Exercise*, *43*(7), 1260–1271. https://doi.org/10.1249/MSS.0b013e3182084929

Simperingham, K. D., Cronin, J. B., Pearson, S. N., & Ross, A. (2019). Reliability of horizontal force–velocity–power profiling during short sprint-running accelerations using radar technology. *Sports Biomechanics*, *18*(1), 88–99. https://doi.org/10.1080/14763141.2017.1386707

Takahashi, K., Kamibayashi, K., Wakahara, T., & Nordez, A. (2020). Muscle size of individual hip extensors in sprint runners: Its relation to spatiotemporal variables and sprint velocity during maximal velocity sprinting. *PLoS One*, *16*(4), e0249670. https://doi.org/10.1371/journal.pone.0249670

Timmins, R. G., Bourne, M. N., Shield, A. J., Williams, M. D., Lorenzen, C., & Opar, D. A. (2016). Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): A prospective cohort study. *British Journal of Sports Medicine*, *50*(24), 1524–1535. https://doi.org/10.1136/bjsports-2015-095362

Timmins, R. G., Opar, D. A., Williams, M. D., Schache, A. G., Dear, N. M., & Shield, A. J. (2014). Reduced biceps femoris myoelectrical activity influences eccentric knee flexor weakness after repeat sprint running: Hamstring EMG and weakness post-running. *Scandinavian Journal of Medicine & Science in Sports*, *24*(4), e299–e305. https://doi.org/10.1111/sms.12171

Van Dyk, N., Behan, F. P., & Whiteley, R. (2019, November). Including the Nordic hamstring exercise in injury prevention programmes halves the rate of hamstring injuries: A systematic review and meta-analysis of 8459 athletes. *British Journal of Sports Medicine*, *53*(21), 1362–1370. https://doi.org/10.1136/bjsports-2018-100045